

Keyword Spotting in Historical Documents

Course: Pattern Recognition - University of Fribourg

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1. Introduction

Keyword Spotting (KWS) is the task of retrieving instances of a given keyword from handwritten document images without requiring full transcription. This is particularly valuable when working with historical manuscripts, where accurate OCR or manual transcription is difficult.

In this project, we implement a **learning-free Keyword Spotting system** using the George Washington Dataset. The system identifies visually similar word images based on their appearance rather than text content. Our approach relies on:

- Extracting word images using provided polygon annotations
- Preprocessing and binarization
- Feature extraction using sliding-window descriptors
- Sequence comparison using **Dynamic Time Warping (DTW)**
- Ranking results based on similarity
- Computing standard KWS evaluation metrics

The goal is to achieve functional retrieval performance using classical pattern recognition techniques and to analyze the results.

2. Methodology

Our approach follows a classical keyword spotting pipeline.

2.1 Preprocessing

For every word:

1. Extract polygon region from the corresponding document image
2. Convert to grayscale
3. Apply **Otsu thresholding** - binarized word image
4. Resize height to fixed value (e.g: 64 px), pad width if needed

2.2 Feature Extraction - Sliding Window Column Features

We slide a **1-pixel column window** across the binarized image and extract a 7-dimensional feature vector per column:

- Mean pixel intensity
- Transition counts
- Upper/lower profile features
- Pixel density features
- Stroke thickness cues

This produces a variable-length sequence per word, depending on word width.

2.3 Similarity Computation using Dynamic Time Warping

Since words vary in width, we use **Dynamic Time Warping (DTW)** to compare two feature sequences. DTW aligns the sequences non-linearly and computes the minimal cumulative cost. Smaller DTW distance more similar words.

We compute DTW distances between:

- Query words (from validation set)
- All candidate words (from training set)

The candidates are ranked by ascending distance.

3. Results and Discussion

3.1 Average Precision (AP)

➤ **AP = 0.2743**

This means that when retrieving all potential matches ranked by DTW distance, the system achieves **27.43% mean average precision**. For a classical keyword spotting system based on handcrafted features, this is a reasonable result.

3.2 Top-K Retrieval Metrics

Top-K	TP	FP	FN	Precision	Recall
Top-1	45	25	401	0.643	0.101
Top-2	78	62	368	0.557	0.175
Top-3	96	114	350	0.457	0.215
Top-5	120	230	326	0.343	0.269

3.3 Interpretation

❖ Top-1 Precision – 0.643

- When showing only the single closest match, the system is correct **64.3%** of the time. This is quite good for noisy historical handwriting.

❖ Recall is low at Top-1 (0.101)

- Only **10.1%** of all relevant keyword occurrences are found immediately. This is typical because handwriting varies significantly.

❖ Increasing K improves recall

- Recall increases from **0.101 - 0.269** when K increases from 1 to 5.
- Precision decreases because more false positives appear.

4. Conclusion

In this project, we implemented a complete classical Keyword Spotting (KWS) system using the George Washington Dataset. Even with a traditional pipeline, based on preprocessing, handcrafted feature extraction, and DTW matching the system demonstrated realistic performance that aligns well with established classical KWS approaches. Several important insights emerged from this work:

- **The sliding-window framework combined with DTW remains a strong and reliable baseline.** Despite its simplicity, it consistently identifies visually similar word patterns and provides competitive precision in ranked retrieval.
- **The low recall values highlight the limitations of handcrafted features**, especially when dealing with large variability in handwriting styles, noise, and subtle word-shape differences. This reinforces why modern systems increasingly rely on learned visual representations.
- **The system performs well when the query word and document words share strong visual similarity**, demonstrating that the pipeline is well-suited for clean, well-defined handwriting instances.

Overall, this implementation serves as a solid starting point for building more advanced keyword spotting systems. It establishes a clear baseline and offers practical insight into the strengths and weaknesses of classical methods. The next natural step is to transition toward **neural KWS approaches**, such as CNN-based embeddings, Siamese networks, or transformer-driven retrieval, which can better capture complex handwriting variations and significantly improve recall.

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